

Topological Group.

Def. Let G be a Group under the binary operation $\cdot: G \times G \rightarrow G$.

A Topological Space $(G, \mathcal{L})$ satisfying T_1 condition, and

$$G \times G \rightarrow G: (x, y) \mapsto xy \text{ and } G \rightarrow G: x \mapsto x^{-1} \text{ are continuous}$$

is called the Topological Group.

Lemma. Let G be a group and T_1 -space. Then

G is Topological Group if and only if $G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Proof. Suppose that G is Topological Group. Then

$\varphi(x, y) = xy$ and $\psi(y) = y^{-1}$ are continuous, therefore

$(x, y) \mapsto xy^{-1} = \varphi(x, \psi(y))$ is continuous, being composition of continuous maps

Conversely, suppose that $\mu: G \times G \rightarrow G: (x, y) \mapsto xy^{-1}$ is continuous.

Then, for fixed $x = 1$, $\mu(1, y) = y^{-1}$ is continuous, so $x \mapsto x^{-1}$ is continuous.

Now, $y \mapsto y^{-1}$ is continuous, so

$(x, y) \mapsto x(y^{-1})^{-1} = xy$ is continuous, being composition of continuous.

~~QED~~

$GL_n(\mathbb{R})$

Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$ equipped Euclidean Topology.

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Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$, with Euclidean space.
Define a Metric for $M_{n \times n}(\mathbb{R})$ as:

$$d: M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}; (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu: M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R}); (A, B) \mapsto \left(\sum_{k=1}^n A_{ik} \cdot B_{ks} \right)_{i,j}$$

is continuous, because it is entry-wise continuous.

Since $GL_n(\mathbb{R})$ is closed under the multiplication, it has also continuous operation.

And, $GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R}); A \mapsto A^{-1}$

is also continuous, because $A^{-1} = (\det A)^{-1} \cdot \text{adj } A$

$$\text{where } (\text{adj } A)_{i,j} = (-1)^{i+j} \cdot \det A(i|j).$$

is also multivariables polynomial, entrywise.

$\sqrt{SL}_n(\mathbb{R})$

Consider $O_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^t = A^{-1}\}$, $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$.

Then, O_n and SO_n are compact, and SO_n is connected.

Proof. $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^n A_{ik} \cdot A_{jk} = \delta_{ij}$.

Define a map $f_{ij}: M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}: A \mapsto \sum_{k=1}^n A_{ik} \cdot A_{jk}$.

Then, it is continuous, therefore

$O_n(\mathbb{R}) = \bigcap_{\substack{1 \leq i \leq n \\ i \neq j}} f_{ij}^{-1}[\{0\}] \cap \bigcap_{1 \leq i \leq n} f_{ii}^{-1}[\{1\}]$ is finite intersection of closed sets,

thus closed. Moreover, $\dim O_n(\mathbb{R}) \leq n$, so it is bounded. Therefore compact.

And, $\det: O_n(\mathbb{R}) \rightarrow \mathbb{R}: A \mapsto \det A$ is continuous, so,

$SO_n(\mathbb{R}) = \det^{-1}[\{1\}]$ also closed, and bounded subset of bounded set.

$\Gamma_{\mathcal{O}_n}(\mathbb{R})$

$\Gamma_{SO_n}(\mathbb{B})$

$$\Gamma_{U_n}(\mathbb{C})$$